A Project Stage-I Report on

AI EQUITY RESEARCH ANALYSIS TOOL

Submitted in Partial fulfillment of requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING

By

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KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY
POWERED BY GENESIS

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTION)

Accredited by NBA & NAAC, Approved by AICTE, Affiliated to JNTUH.

Narayanaguda, Hyderabad, Telangana-29

2025-26

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that this is a bonafide record of the project report titled **"AI EQUITY RESEARCH ANALYSIS TOOL"** which is being presented as the **Project Stage-I** report in **IV year I Semester (RKR21)** by

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In partial fulfillment for the award of the degree of Bachelor of Technology in Computer Science and Engineering at Keshav Memorial Institute of Technology (An Autonomous Institution) Narayanaguda, affiliated to the Jawaharlal Nehru Technological University Hyderabad, Hyderabad

Faculty Supervisor	Senior Faculty Member	Head of Department
(Ms Spoorthy H)	(Ms Premlatha)	(Mr. Para Upendar)

Vision & Mission of The Institute

Vision of the Institution

- To be the fountain head in producing highly skilled, globally competent engineers.
- Producing quality graduates trained in the latest software technologies and related tools and striving to make India a world leader in software products and services.

Mission of the Institution

- To provide a learning environment that inculcates problem solving skills, professional, ethical responsibilities, lifelong learning through multi modal platforms and prepares students to become successful professionals.
- To establish an industry institute Interaction to make students ready for the industry.
- To provide exposure to students on the latest hardware and software tools.
- To promote research-based projects/activities in the emerging areas of technology convergence.
- To encourage and enable students to not merely seek jobs from the industry but also to create new enterprises.
- To induce a spirit of nationalism which will enable the student to develop, understand India's challenges and to encourage them to develop effective solutions.
- To support the faculty to accelerate their learning curve to deliver excellent service to students.

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Vision of the Department

To be among the region's premier teaching and research Computer Science and Engineering departments producing globally competent and socially responsible graduates in the most conducive academic environment.

Mission of the Department

- To provide faculty with state of the art facilities for continuous professional development and research, both in foundational aspects and of relevance to emerging computing trends.
- To impart skills that transform students to develop technical solutions for societal needs and inculcate entrepreneurial talents.
- To inculcate an ability in students to pursue the advancement of knowledge in various specializations of Computer Science and Engineering and make them industry-ready.
- To engage in collaborative research with academia and industry and generate adequate resources for research activities for seamless transfer of knowledge resulting in sponsored projects and consultancy.
- To cultivate responsibility through sharing of knowledge and innovative computing solutions that benefit the society-at-large.
- To collaborate with academia, industry and community to set high standards in academic excellence and in fulfilling societal responsibilities

PROGRAM OUTCOMES (POs)

PO1. Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2. Problem Analysis: Identify formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences

PO3. Design/Development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4. Conduct Investigations of Complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5. Modern Tool Usage: Create select, and, apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6. The Engineer and Society: Apply reasoning informed by contextual knowledge to societal, health, safety. Legal und cultural issues and the consequent responsibilities relevant to professional engineering practice.

PO7. Environment and Sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of, and need for sustainable development.

PO8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9. Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse teams and in multidisciplinary settings.

PO10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11. Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12. Life-Long Learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1: An ability to analyze the common business functions to design and develop appropriate Information Technology solutions for social upliftments.

PSO2: Shall have expertise on the evolving technologies like Python, Machine Learning, Deep learning, IOT, Data Science, Full stack development, Social Networks, Cyber Security, Mobile Apps, CRM, ERP, Big Data, etc.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO1: To imbibe analytical and professional skills for successful careers and create enthusiasts to pursue advance education supplementing their career growth.

PEO2: Graduates will solve real time problems design, develop and implement innovative ideas by applying their computer engineering principles.

PEO3: Graduates will develop necessary skillset for industry by imparting state of art technology in various areas of computer science engineering.

PEO4: Graduates will engage in lifelong learning and be able to work collaboratively exhibiting high level of professionalism.

DECLARATION

We hereby declare that the results embodied in the dissertation entitled **"AI EQUITY RESEARCH ANALYSIS TOOL"** has been carried out by us together during the academic year 2025-26 as a partial fulfillment of the award of the B.Tech degree in Computer Science and Engineering from Keshav Memorial Institute of Technology (An Autonomous Institution) Narayanaguda, affiliated to JNTUH. We have not submitted this report to any other university or organization for the award of any other degree.

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We would like to thank the entire CSE Department faculty, who helped us directly and indirectly in the completion of the project.

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ABSTRACT

In this project, a prototype and implementation of an AI-powered Equity Research Analysis Tool is demonstrated. The proposed system consists of a Retrieval-Augmented Generation (RAG) backend and an interactive user interface. The backend interface includes FastAPI, Pinecone, and Groq, Llama 3.1, while the software interface includes a React-based dashboard for visualization. We automate the extraction of financial data using hybrid scraping techniques involving Trafilatura and Playwright to handle static and dynamic content. A Generative AI pipeline is used for cleaning unstructured text, generating embeddings, and producing executive summaries. An application is provided for managing URL sources and performing natural language queries via a chatbot. This system is one of the best methods for accelerating financial research with high accuracy and one of the best methods for an automated insight generation system. This system is also expandable for audio-based inputs using speech-to-text integration and real-time market data analysis.

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CHAPTER-1

INTRODUCTION

1.1 Purpose of the Project

The purpose of the AI Equity Research Analysis Tool is to provide a smart assistant for equity research and financial data analysis by automating the processing of finance-related web content. The system is designed to accept multiple publicly accessible URLs such as news articles, company pages, and stock analysis reports, and convert them into structured insights that analysts can use in their daily workflows. It aims to reduce the time spent on manual reading, enable faster identification of key metrics and risk factors, and support more informed investment decisions through AI-driven summaries and question answering.

1.2 Problem with Existing Systems

Existing equity research workflows are heavily manual and depend on analysts reading large numbers of articles, reports, and web pages individually. This leads to a slow and repetitive research process, where valuable time is spent copying information into spreadsheets or notes instead of performing higher-level analysis. There is no unified analysis across multiple sources, and analysts must rely on static tools such as PDFs and Excel sheets without any conversational interface or automated summarization of the collected data. Current systems also lack intelligent mechanisms to extract insights, combine information from multiple web sources, and provide interactive Q&A grounded in the article content.

1.3 Proposed System

The proposed system is an AI-powered, RAG-based equity research tool that automates the entire pipeline from URL ingestion to interactive analysis. Users provide finance-related URLs through a React-based frontend, after which the backend uses Playwright and Beautiful Soup to scrape and clean the textual content from both static and JavaScript-heavy websites. The cleaned content is converted into vector embeddings using HuggingFace models and stored in Pinecone, enabling semantic search and retrieval. A Large Language Model, integrated through a RAG pipeline, generates concise summaries and answers user questions by using only the retrieved document content as context. This results in a unified, structured research platform that supports automated summarization, multi-URL analysis, and conversational querying.

1.4 Scope of the Project

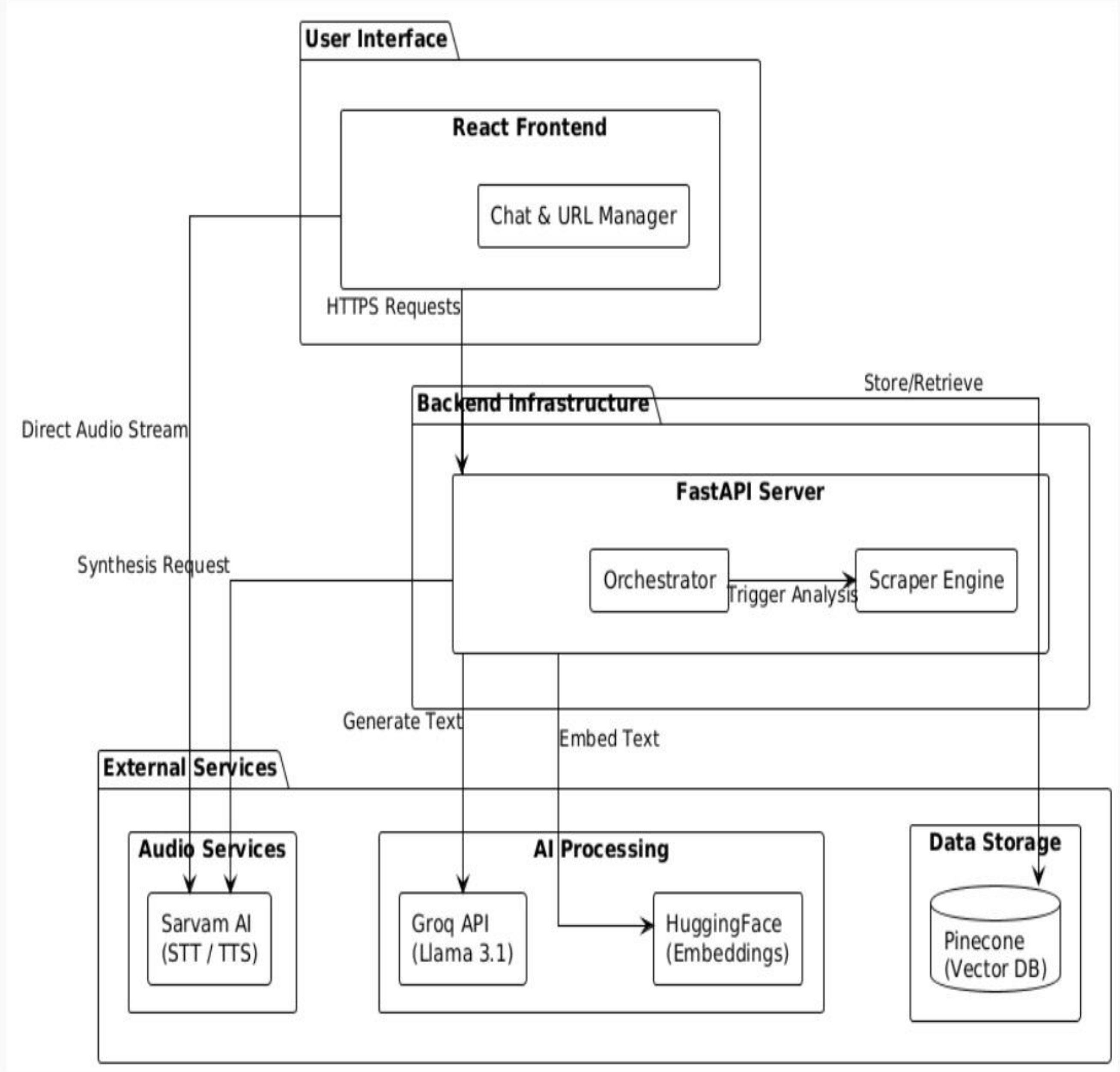
The scope of this project includes the design and implementation of all key components required for an industry-oriented equity research assistant. It covers frontend development for URL input, summary display, and chatbot interaction; backend APIs for scraping, extraction, embedding, and retrieval; integration with Pinecone for vector storage; and deployment of the RAG-based AI layer for summarization and Q&A. The system focuses on publicly accessible finance websites such as Moneycontrol, Yahoo Finance, Zerodha, and other market news or analysis pages. The project does not address handling of private data, live trading, or portfolio management, but creates a solid foundation that can be extended with additional financial analytics and features in later stages.

1.5 Architecture Diagram (Description)

The overall architecture follows a multi-layer design with clear separation between frontend, backend, data, and AI layers.

- At the frontend layer, a React.js application allows users to submit URLs, monitor processing, view summaries, and chat with the assistant.
- The backend layer, implemented in FastAPI, exposes endpoints that trigger Playwright-based scraping, Beautiful Soup parsing, and embedding generation for the extracted text.

- The data layer uses Pinecone as the vector database to store document embeddings and support semantic similarity queries.
- The AI layer retrieves the most relevant document chunks from Pinecone and passes them, along with user queries, to an LLM deployed on Hugging Face Spaces, forming a RAG pipeline that produces grounded summaries and answers.
  Deployment is handled using Vercel for the frontend and platforms such as Render or Hugging Face Spaces for backend and AI services, with GitHub used for version control and collaboration.


User Interface
React Frontend
Chat & URL Manager
HTTPS Requests
Direct Audio Stream
Store/Retrieve
Backend Infrastructure
FastAPI Server
Synthesis Request
Orchestrator
Trigger Analysis
Scraper Engine
Generate Text
Embed Text
External Services
Audio Services
AI Processing
Data Storage
Sarvam AI
(STT / TTS)
Groq API
(Llama 3.1)
HuggingFace
(Embeddings)
Pinecone
(Vector DB)

CHAPTER-2

LITERATURE SURVEY

Equity research has traditionally depended on manual analysis of financial reports, news articles, earnings call transcripts, and market commentary, which makes the process time-consuming and difficult to scale. Early computational tools in finance primarily focused on numerical time-series analysis and basic statistical models, with limited ability to interpret unstructured text that carries a large portion of market-relevant information. As a result, analysts often relied on personal judgment and ad-hoc note taking, leading to inconsistent coverage and potential information overload when tracking many companies or sectors simultaneously.

With the rise of Natural Language Processing (NLP), researchers began to apply techniques such as bag-of-words, TF–IDF, and classical machine learning classifiers for sentiment analysis of financial news and firm-specific disclosures. These methods allowed basic polarity detection (positive/negative) and topic identification, which could be correlated with stock price movements or volatility. However, they struggled with financial domain nuances, context dependence, and long-range dependencies in text, limiting the quality of insights for complex equity research tasks such as risk assessment or multi-document comparison.

The introduction of deep learning and, in particular, transformer-based architectures such as BERT and its financial-domain variants, significantly improved the ability to capture context and semantics in financial language. Pretrained language models fine-tuned on financial corpora have been used for tasks like earnings call sentiment classification, event extraction, and automated report generation, achieving better performance than earlier statistical methods. Despite these advances, many systems remained task-specific and operated on curated datasets rather than arbitrary, user-provided web content.

In parallel, the concept of vector representations (embeddings) for sentences and documents, along with specialized vector databases, opened up new possibilities for semantic search over large collections of unstructured text. Instead of simple keyword matching, embeddings allow retrieval based on meaning, which is valuable when analysts search for similar events, risk factors, or themes across numerous financial articles and company updates. Managed vector databases such as Pinecone provide indexing, similarity search, and scalability features that make it feasible to build real-time semantic retrieval systems on top of web-scale text.

More recently, Retrieval-Augmented Generation (RAG) has emerged as a powerful pattern that combines vector retrieval with large language models. In a RAG system, user queries are first matched against a vector store to retrieve the most relevant document chunks, which are then supplied as context to an LLM that generates grounded, context-aware answers. This architecture reduces hallucinations, keeps responses tied to source documents, and enables dynamic, domain-specific assistants without retraining the base model.

In the finance domain, some tools and prototypes now integrate news feeds, research reports, and company filings with LLM-based summarization and Q&A. However, many of these are closed, proprietary platforms or tightly bound to specific data providers, and they often do not allow analysts to flexibly supply their own URLs from diverse sites such as Moneycontrol, Yahoo Finance, Zerodha, and other public web sources. There is still a gap for systems that can robustly scrape JavaScript-heavy sites, clean and normalize the content, build semantic indices, and expose a transparent RAG pipeline that analysts can use as a personal research assistant.

The AI Equity Research Analysis Tool in this project is positioned within this evolving landscape. It brings together web scraping (Playwright, Beautiful Soup), embedding-based semantic search (HuggingFace + Pinecone), and RAG-driven summarization and question answering, specifically tailored for equity research tasks. By supporting arbitrary finance-related URLs and providing a conversational interface on top of a custom, user-controlled knowledge base, the system extends current literature trends into a practical, deployable assistant for analysts, investors, and students.

CHAPTER-3

SOFTWARE REQUIREMENT SPECIFICATION

3.1 Introduction to SRS

The Software Requirement Specification (SRS) defines the complete set of requirements for the AI Equity Research Analysis Tool, covering functional behavior, constraints, and system interfaces. It acts as a reference for design, development, and testing, ensuring that the final implementation satisfies the expectations of stakeholders and aligns with the objectives of an industry-oriented major project.

3.2 Role of SRS

The role of the SRS is to provide a clear, unambiguous description of what the system must do before any detailed design or coding begins. It serves as a contract between the project team and the guide/examiners, supports planning of milestones, and enables traceability from requirements to design elements and test cases. A well-defined SRS helps avoid misunderstandings, reduces rework, and improves overall project quality.

3.3 Requirements Specification Document

The requirements specification document is organized into sections describing functional requirements, non-functional requirements, performance constraints, and hardware/software needs, in line with the IQAC template. For this project, the document focuses on requirements related to URL input management, web scraping, embedding generation, vector storage, RAG-based summarization, and conversational question answering for equity research.

3.4 Functional Requirements

The key functional requirements are listed below.

- URL Input and Management
    - The system shall provide an interface for users to add one or more finance-related URLs.
    - The system shall allow users to view and manage the list of URLs to be processed.
- Web Content Extraction
    - The system shall fetch and render static and JavaScript-heavy websites using Playwright.
    - The system shall clean and extract main article content from the fetched HTML using Beautiful Soup.
- Embedding and Vector Storage
    - The system shall generate embeddings for extracted text using HuggingFace models.

- The system shall store embeddings and associated metadata in Pinecone to enable semantic retrieval.

- AI Summary Generation
  - The system shall generate concise summaries that highlight important points and key metrics from the processed URLs.
  - The system shall display summaries in the frontend in a readable format.
- Conversational Question Answering
  - The system shall provide a chatbot interface where users can ask equity-related questions.
  - For each question, the system shall retrieve relevant document chunks from Pinecone and use an LLM to generate answers grounded in those chunks.
- Error Handling and Notifications
  - The system shall notify users when a URL is invalid, unreachable, or fails to process.
  - The system shall provide appropriate error messages and options to retry or modify inputs.

3.5 Non-Functional Requirements

The non-functional requirements of the system include the following.

- Performance:
  - Typical URL processing (scraping, cleaning, embedding, and summary generation) should complete within approximately 5–10 seconds under normal conditions.
- Scalability:
  - The system should handle multiple URLs and large article content sizes without significant performance degradation.
- Reliability:
  - The RAG pipeline should consistently return relevant and accurate information based on stored documents, minimizing incorrect or hallucinated outputs.
- Security:
  - The system must handle API keys and configuration values securely and should not store or expose sensitive user data.
- Usability:
  - The interface should be intuitive for target users such as analysts and students, with clear navigation for submitting URLs, viewing results, and chatting with the assistant.

3.6 Performance Requirements

The system must maintain responsive behavior even when processing multiple URLs. Each URL should be scraped and embedded within a short time window, and once documents are stored, subsequent queries to the chatbot should retrieve results and generate responses nearly in real time. The design should also consider external API rate limits and ensure graceful degradation or queuing when limits are reached.

3.7 Software Requirements

The software requirements for implementation are:

- Frontend:
  - React.js, HTML, CSS
- Backend:
  - Python 3.x, FastAPI
- Web Scraping:
  - Playwright for rendering and capturing dynamic pages
  - Beautiful Soup for parsing and extracting article text
- AI and Data:
  - HuggingFace embeddings and LLM models
  - Pinecone vector database
- Deployment and Tools:
  - Vercel for frontend deployment
  - Render and/or Hugging Face Spaces for backend and model hosting
  - Git and GitHub for version control and collaboration

3.8 Hardware Requirements

For development and testing, the system requires a machine with a modern multi-core CPU, at least 8–16 GB of RAM, and stable internet connectivity to interact with web pages and external AI services. For production deployment, cloud instances with scalable compute and memory resources are recommended to handle multiple concurrent scraping and query operations, as well as the overhead of embedding generation and LLM inference.

CHAPTER-4

4.SYSTEM DESIGN

4.1 Introduction to UML

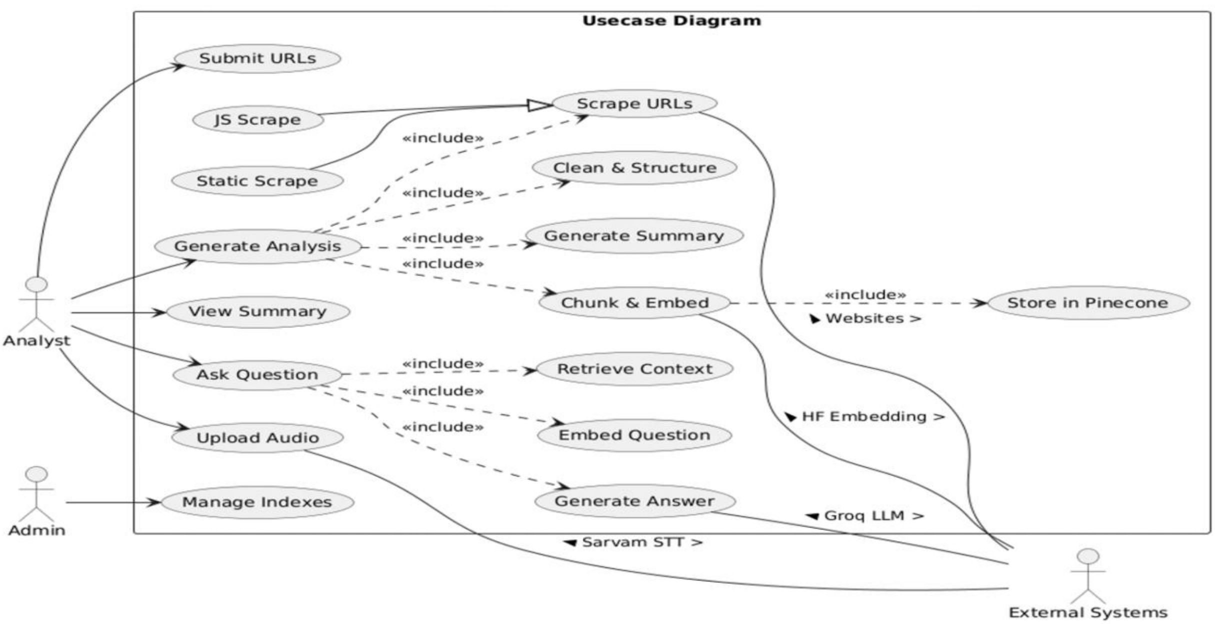
Unified Modeling Language (UML) is used to capture the structural and behavioral aspects of the AI Equity Research Analysis Tool before implementation. UML diagrams such as use case, sequence, class, activity, and deployment diagrams help visualize how users interact with the system, how components collaborate, and how data moves through the architecture.

4.2 UML Diagrams

This project uses multiple UML diagrams to represent different perspectives of system design, including use case, sequence, state/activity, deployment, and architecture diagrams. These diagrams collectively describe the behavior of the system from user interaction down to backend services and infrastructure-level deployment.

4.2.1 Use Case Diagram

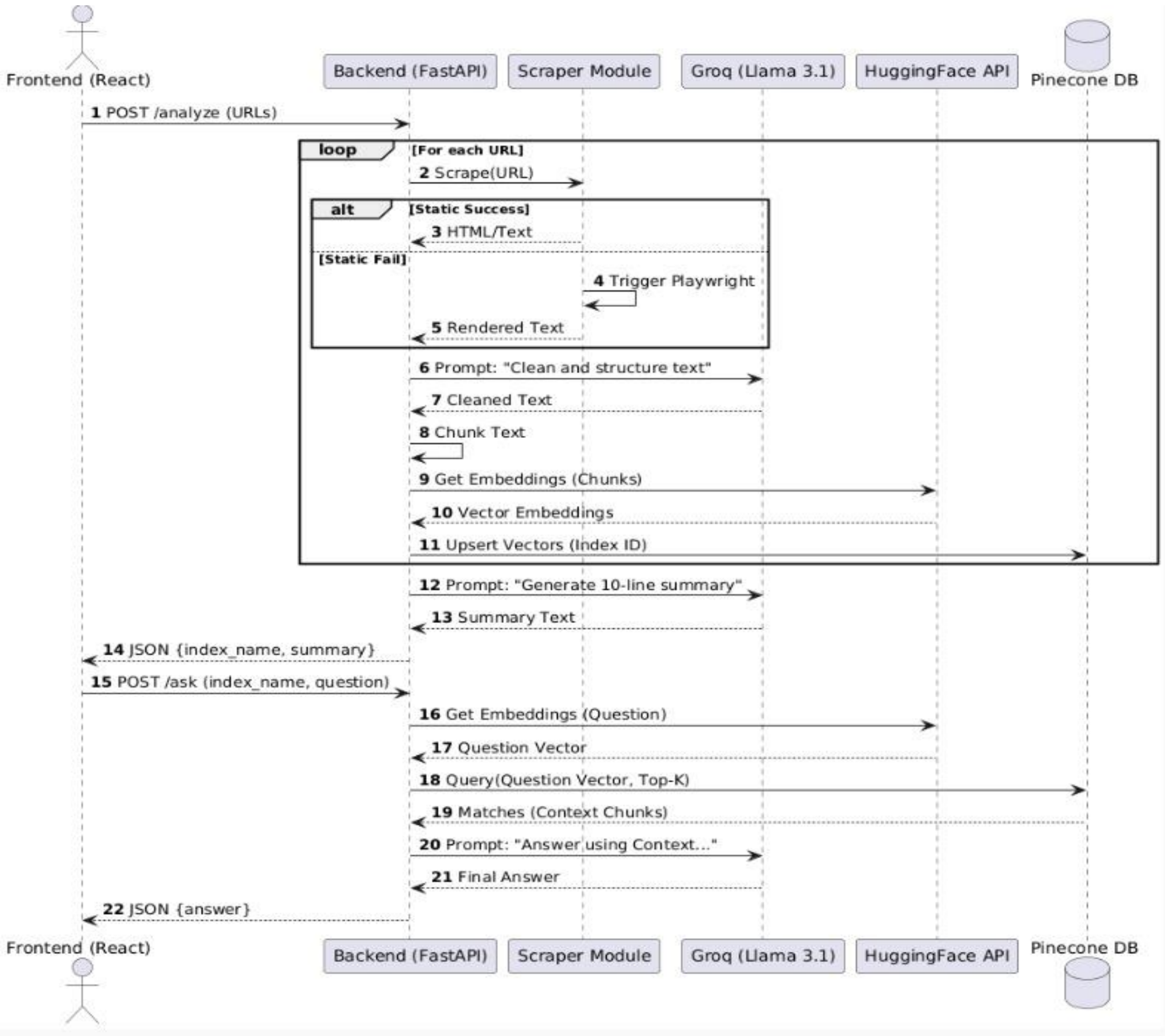
The use case diagram identifies the primary actor as the “Equity Research User,” which includes roles such as analysts, investors, and students. Main use cases are: “Submit URLs,” “View Processed Articles,” “View Summaries,” and “Ask Questions via Chatbot.” The system boundary encloses backend services like scraping, embedding, and RAG-based response generation, which are triggered by user actions from the frontend.



4.2.2 Sequence Diagram

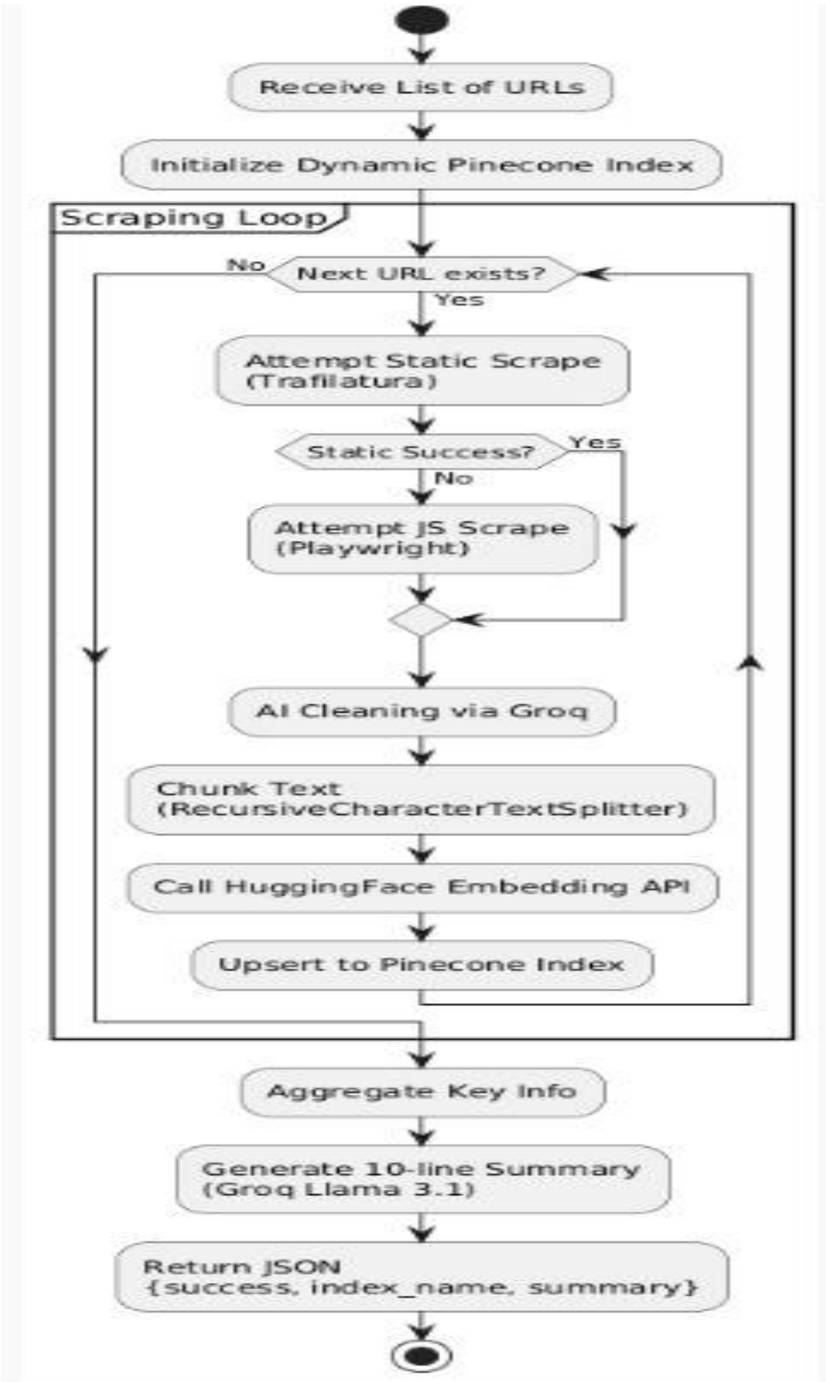
The sequence diagram shows the dynamic interaction for two key flows: URL processing and question answering.

- In the URL processing flow, the user sends URLs from the React frontend, the FastAPI backend receives them, invokes Playwright for scraping and Beautiful Soup for parsing, calls the embedding service to generate vectors, and stores them in Pinecone.
- In the question answering flow, the user sends a query to the chatbot, the backend retrieves similar embeddings from Pinecone, passes the retrieved context to the LLM, and sends the generated answer back to the frontend.



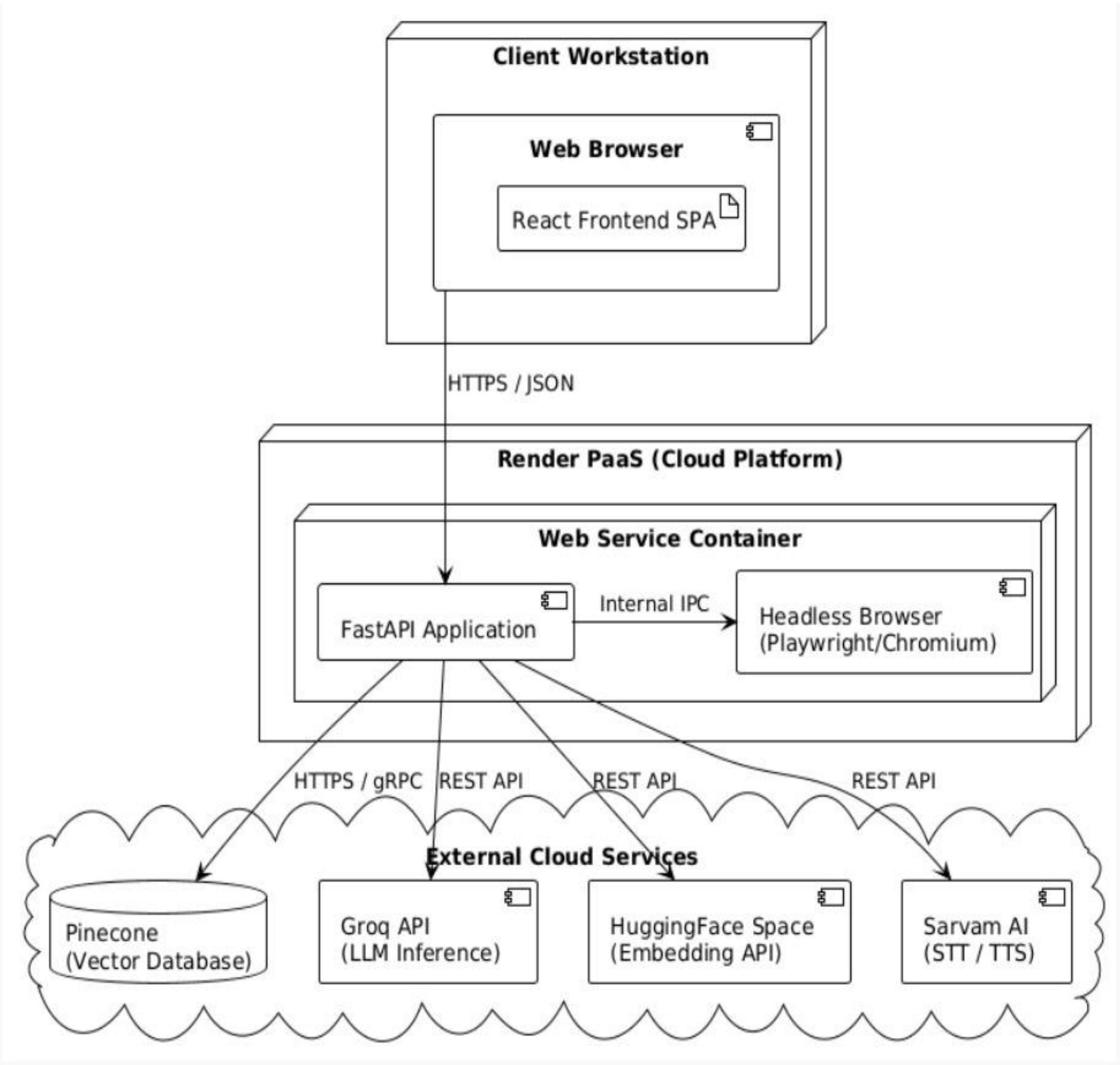
4.2.3 State / Activity Diagram

The state or activity diagram describes the life cycle of a URL from the moment it is submitted until it is fully processed and available for querying. Typical states include "Received," "Scraping," "Parsing," "Embedding," "Indexed," "Failed," and "Completed." The activity flow covers branching for success or failure at each step, along with transitions into user-facing states like "Ready for Summary" and "Ready for Q&A."



4.2.4 Deployment Diagram

The deployment diagram illustrates how different software components are mapped onto physical or virtual infrastructure. It includes nodes such as the user device (browser), frontend server (Vercel), backend API server (Render or similar), vector database service (Pinecone), and AI model hosting (Hugging Face Spaces). Communication links represent HTTPS calls between browser and frontend, API calls between frontend and backend, and service calls from backend to Pinecone and Hugging Face.



4.3 Technologies Used

The technologies used in this system are organized by layer.

- Frontend: React.js and CSS for building the URL input interface, displaying real-time summaries, and providing a chatbot UI.
- Web Scraping: Playwright to handle JS-heavy finance sites and Beautiful Soup for HTML parsing.
- Backend: FastAPI with Python to orchestrate scraping, extraction, embedding, and RAG workflow.
- AI Layer: HuggingFace embeddings, Pinecone vector database, and an LLM for summary and Q&A.
- Deployment: Vercel for frontend hosting, Render/Hugging Face Spaces for backend and AI services, and GitHub for version control.